

Final Assignment

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So your team becomes:

Person	Role	Owns
P1	Frontend	Web app + GIS visualization
P2	Backend	API + PostGIS + integration
P3	Data	Data sources + preprocessing + GIS data
P4	ML	Hazard AI + risk + priority + AHP
P5	UI/UX	User flow + Figma + design system
P6	Research/Testing & Deploy	Research + validation + testing + Deploy

P1 : Frontend

Person 1 — Frontend Developer

Own this entire area

Web application + interactive GIS interface

Responsibilities

- Next.js setup
- TypeScript
- Dashboard
- Interactive map
- Risk visualization
- Village/settlement details
- Population-at-risk screens
- Relocation recommendation screen
- Candidate-site comparison
- API integration
- Loading/error states

Tech stack

Next.js

TypeScript

Tailwind CSS

MapLibre GL JS

Deck.gl

Recharts

Main screens

1. Dashboard
2. Risk Map
3. Hazard Analysis
4. Vulnerable Habitations
5. Population at Risk
6. Relocation Priority
7. Safe-Site Discovery
8. Site Comparison
9. Recommended Site
10. Reports

Person 1's final deliverable

A working website that consumes backend APIs.

P2 : Backend

Person 2 — Backend Developer

This person should be the **integration owner**.

This is extremely important.

Responsibilities

Build the central backend that connects:

Frontend



Backend



ML



GIS/Data



Database

Tech stack

Python

FastAPI

PostgreSQL

PostGIS

Docker

APIs

Create APIs such as:

GET /api/hazards

GET /api/risk-map

GET /api/habitations

GET /api/population-risk

GET /api/relocation-priority

GET /api/candidate-sites

GET /api/site-ranking

GET /api/carrying-capacity

GET /api/recommendation

Database

Person 2 owns the schema for:

villages
hazards
risk_zones
buildings
roads
population
facilities
candidate_sites
relocation_scores
site_suitability
carrying_capacity

Most important responsibility

Define the **API/data contracts** between everyone.

For example:

```
{  
  "village_id": "V001",  
  "name": "Village A",  
  "risk_score": 0.87,  
  "risk_level": "RED",  
  "population": 3240,  
  "population_at_risk": 2810,  
  "relocation_priority": "IMMEDIATE"  
}
```

Person 1 should be able to build the frontend using this contract even while Person 4 is still developing the ML model.

P3 : Data Sources & Preprocessing

Person 3 — Data Sources & Processing

This person is responsible for the **entire data pipeline**.

This is arguably one of the most important roles for your project because your ML model is only as good as the data.

Responsibilities

Data collection

Collect:

DEM / Elevation
Rainfall
Historical disasters
Land use / land cover
Rivers
Roads
Villages
Population
Buildings
Schools
Hospitals
Water resources

Potential sources:

Google Earth Engine
OpenStreetMap
DEM datasets
Population datasets
Historical disaster datasets
Government/open datasets

Processing

Raw Data

↓

Cleaning

↓

Coordinate normalization

↓

Missing-value handling



Spatial processing



Feature extraction



ML-ready dataset

GIS work

This person should also handle:

- GeoPandas
- Rasterio
- GDAL
- Shapely
- Coordinate systems
- Raster/vector conversion
- Spatial joins
- DEM → slope
- Distance calculations

Output

For example:

village_id
latitude
longitude
elevation
slope
rainfall
distance_to_river
land_use
population
households
historical_flood_count
historical_landslide_count
...

This becomes the **input to Person 4's ML model**.

P4 : ML

Person 4 — ML Engineer

Person 4 owns the **AI/decision intelligence**.

Don't give them only "train an ML model." Their work should cover the entire intelligence pipeline.

Module A — Hazard Prediction

Start with **flood prediction**.

Rainfall
Elevation
Slope
River distance
Land use
Historical flood data

↓

XGBoost

↓

Flood probability

Then expand to:

Flood
Landslide
Cloudburst
Coastal erosion

Module B — Multi-Hazard Risk

Combine individual hazard probabilities:

Flood Risk

+

Landslide Risk

+

Cloudburst Risk

+

Erosion Risk

↓
Multi-Hazard Risk Score
↓
RED / ORANGE / YELLOW / GREEN

Module C — Relocation Priority

Create:

Relocation Priority Index

Inputs:

Hazard severity
Population exposure
Population vulnerability
Buildings
Critical facilities
Historical disasters

Output:

IMMEDIATE
SHORT-TERM
MEDIUM-TERM
MONITOR

Module D — AHP

Person 4 can implement the mathematical side of AHP:

Safety
Road access
Water
Healthcare
Education
Land availability
Electricity
Environment

↓
AHP
↓

Module E — Carrying Capacity

Calculate:

Land capacity

Water capacity

Healthcare capacity

Education capacity

Infrastructure capacity



Final carrying capacity

Tech stack

Python

Pandas

NumPy

Scikit-learn

XGBoost

GeoPandas

OR-Tools

PyTorch can be added later if you actually need deep learning.

P5 : UI/Ux

Person 5 — UI/UX Designer

This person should **not wait until the frontend is finished**.

They should design the complete user journey first.

Main goal

Make the system understandable to a **government/administrative decision-maker**, not a GIS expert.

Design

Dashboard

Disaster Risk Dashboard				
Critical	High	Moderate	Low	
42	87	213	903	
RISK MAP				
Population at Risk 1.24M				
Immediate Relocation 27 Villages				

Important screens

Design:

1. Dashboard
2. Multi-hazard map
3. Village details
4. Population exposure
5. Relocation priority
6. Candidate relocation sites
7. Site comparison
8. Carrying capacity
9. Final recommendation

10. Report

Person 5 should deliver

Figma



Design System



Wireframes



High-fidelity UI



Component specifications



Frontend handoff

P6 : Research & Testing

Person 6 — Research & Testing

This person should **not be treated as just documentation**.

Give them ownership of **validation + research + system evaluation**.

Responsibilities

1. Research

Research:

- Existing disaster-management systems
 - Existing hazard maps
 - Existing relocation frameworks
 - Existing GIS solutions
 - Existing AI disaster prediction approaches
 - Existing AHP approaches
 - Government guidelines
 - Relevant research papers
-

2. Dataset validation

Check:

Is the dataset reliable?
Are coordinates correct?
Are values missing?
Are units consistent?
Is the geographic coverage correct?

3. ML validation

Person 6 works with Person 4 to evaluate:

Accuracy
Precision
Recall
F1-score

ROC-AUC
Confusion Matrix

For hazard prediction, **don't just report accuracy**. In imbalanced disaster datasets, precision/recall and calibration can be more informative.

4. GIS validation

Test:

Does the red zone correctly intersect villages?

Are affected populations calculated correctly?

Are candidate sites actually outside hazardous areas?

Are distances calculated correctly?

5. Relocation validation

Test scenarios such as:

Scenario 1

High hazard + high population

→ Immediate

Scenario 2

Moderate hazard + low population

→ Medium-term

Scenario 3

Low hazard

→ Monitor

6. End-to-end testing

They should test:

Data

↓

ML

